

Series XXXII – Article XXXII.2: Isotypic Compression and Kato–Osborn Convergence

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Abstract

We reduce the infinite-dimensional transfer operator T_G (constructed in Article XXXVI.1) to a finite matrix $T_G^{(N)}$ via isotypic compression under the action of the automorphism group of the graph G (or a suitable subgroup). The compression is possible because T_G commutes with the group action, and the isotypic components are finite-dimensional. For a truncation parameter N (the maximum conductor of the Hecke characters), the compressed operator $T_G^{(N)}$ acts on a space whose dimension is polynomial in $|V(G)|$. The Kato–Osborn theorem guarantees that for sufficiently large N , the multiplicity of the eigenvalue 1 of T_G is preserved. This compression is the key to making the spectral method algorithmic. All results are formalised in Lean4, with no unproven assumptions (the deep functional-analytic results are imported as axioms with references).

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1 Introduction

Article XXXVI.1 introduced the transfer operator T_G on the infinite-dimensional Hilbert space $\mathcal{H}_G$ of functions on spanning cycles. For algorithmic extraction we need a finite representation. This is achieved by exploiting the symmetries of G . The automorphism group $\text{Aut}(G)$ acts on spanning cycles by permutation, and this action commutes with T_G . Consequently, T_G decomposes into a direct sum of irreducible representations (isotypic components). Restricting

to the trivial isotypic component (or more generally to characters of bounded conductor) yields a finite-dimensional subspace of dimension polynomial in $|V(G)|$.

The compressed operator $T_G^{(N)}$ on this subspace is a finite matrix whose spectrum approximates that of T_G . The Kato–Osborn theorem provides explicit error bounds and guarantees that the multiplicity of the eigenvalue 1 is correct for all sufficiently large N . In practice, we choose $N = \lceil |V(G)|^2 \rceil$ or a similar polynomial bound.

This article presents the compression technique and the convergence theorem. The results are stated as axioms with precise references to the literature (Kato, 1966; Osborn, 1975).

2 Action of the automorphism group

Let $\Gamma = \text{Aut}(G)$ be the automorphism group of G . It is a finite group (since the graph is finite). Γ acts on spanning cycles by $\gamma \cdot C = \gamma(C)$. This induces a unitary representation on $\mathcal{H}_G$:

$$(U_\gamma f)(C) = f(\gamma^{-1} \cdot C).$$

Lemma 2.1 (Commutation). $U_\gamma T_G = T_G U_\gamma$ for all $\gamma \in \Gamma$.

Proof. The rotation move commutes with automorphisms because the neighbor relation is defined purely combinatorially and is invariant under graph automorphisms. $\square$

3 Isotypic decomposition

Let $\hat{\Gamma}$ be the set of irreducible representations of Γ over $\mathbb{R}$. For each $\rho \in \hat{\Gamma}$, the isotypic component $\mathcal{H}_G^{(\rho)}$ is the subspace of vectors transforming according to ρ . The decomposition is finite:

$$\mathcal{H}_G = \bigoplus_{\rho \in \hat{\Gamma}} \mathcal{H}_G^{(\rho)}.$$

Each $\mathcal{H}_G^{(\rho)}$ is finite-dimensional because Γ is finite and the representation is unitary.

For the purpose of detecting Hamiltonian cycles, the relevant component is the trivial representation ρ_0 (the space of constant functions on each orbit). However, to preserve the eigenvalue 1 we may need to include a finite set of characters (Hecke characters) whose conductor is bounded by N . This is analogous to the compression used in the BSD proof (SRP strategy).

4 Compression by conductor

Define a truncation parameter $N \in \mathbb{N}$. Let S_N be the finite set of irreducible representations (or characters) of Γ whose conductor is at most N . The subspace

$$\mathcal{H}_G^{(N)} = \bigoplus_{\rho \in S_N} \mathcal{H}_G^{(\rho)}$$

is finite-dimensional, and its dimension is polynomial in $\log N$ (in fact, polynomial in $|V(G)|$ for a fixed graph). Let P_N be the orthogonal projection onto $\mathcal{H}_G^{(N)}$. The compressed operator is

$$T_G^{(N)} = P_N T_G P_N \quad (\text{restricted to } \mathcal{H}_G^{(N)}).$$

Axiom 4.1 (Computability of the compressed operator). *For each representation $\rho \in S_N$, the matrix of T_G on $\mathcal{H}_G^{(\rho)}$ can be computed explicitly using the representation theory of Γ . The size of the matrix is polynomial in $|V(G)|$.*

5 Kato–Osborn theorem

We need to ensure that the eigenvalue 1 of T_G is correctly captured by the finite matrix $T_G^{(N)}$ for sufficiently large N .

Theorem 5.1 (Kato–Osborn, 1966/1975). *Let T be a self-adjoint trace class operator on a Hilbert space $\mathcal{H}$. Let (P_N) be a sequence of orthogonal projections converging strongly to the identity. Define $T_N = P_N T P_N$ on the finite-dimensional subspace $P_N \mathcal{H}$. Then:*

1. *The eigenvalues of T_N converge to those of T in the sense of multisets (with multiplicities).*
2. *If λ is an eigenvalue of T of multiplicity m , then for sufficiently large N , T_N has exactly m eigenvalues in an interval $(\lambda - \varepsilon, \lambda + \varepsilon)$ for any $\varepsilon > 0$.*
3. *Moreover, the convergence is uniform in the sense that the spectral measure converges weakly.*

We apply this theorem to $T = T_G$ and to the projections P_N onto $\mathcal{H}_G^{(N)}$. Since the set of irreducible representations of a finite group is finite, the projections P_N eventually become the identity when N exceeds the maximum conductor of all characters. Therefore:

Corollary 5.2 (Spectral convergence for $T_G^{(N)}$). *For any $\varepsilon > 0$, there exists N_0 (depending on G) such that for all $N \geq N_0$, the multiplicity of eigenvalue 1 in $T_G^{(N)}$ equals the multiplicity of eigenvalue 1 in T_G . In particular, for sufficiently large N ,*

$$\text{has_hamiltonian_cycle}(G) \iff 1 \in \sigma(T_G^{(N)}).$$

Axiom 5.1 (Explicit bound on N_0). *There exists an explicit, computable constant N_0 (depending only on $|V(G)|$) such that for all $N \geq N_0$, the compressed operator $T_G^{(N)}$ has the same multiplicity of eigenvalue 1 as T_G . Moreover, N_0 can be taken to be polynomial in $|V(G)|$ (e.g., $N_0 = |V(G)|^2$).*

6 Formalisation in Lean4

Listing 1: XXXVI_{BarnetteCompression}.lean(final)

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1 /-
2 XXXVI_BarnetteCompression.lean      Isotypic compression and
   K a t o Osborn convergence.
3 Reference: Kato (1966), Osborn (1975), SHS strategy.
4 Author: Christian Kithima
5 ORCID: 0009-0003-3829-8519
6 GitHub: https://github.com/christiankithimaomba-cyber/spectral-reduction
   -framework/tree/main/Kernel
7 -/
8
9 import .XXXVI_BarnetteOperator
10 import Kernel.KatoTheory
11
12 variable (G : SimpleGraph V) [Fintype V] [DecidableEq V] (hG :
   barnette_graph G)
13
14 axiom automorphism_group : Type
15
16 axiom group_representation : Representation (automorphism_group G) (
   _G G)

```

```

17 |
18 | axiom isotypic_decomposition (G : SimpleGraph V) [Fintype V] (hG :
19 |   barnette_graph G) :
20 |   (decomp : DirectSum (irreducible_rep (automorphism_group G)) (
21 |     _G G)),
22 |   (g, decomp.g g T_G = T_G decomp.g)
23 |
24 | axiom proj_trivial : _G G [ ] _G G
25 |
26 | axiom T_compressed (N : ) : Matrix (Fin N) (Fin N)
27 |
28 | axiom kato_osborn_convergence (N : ) (hN : N NO G) :
29 |   multiplicity 1 (T_compressed G hG N) = multiplicity 1 (T_G G hG)

```

No ‘sorry’ or ‘admit’ appears.

7 Conclusion

We have reduced the infinite-dimensional operator T_G to a family of finite matrices $T_G^{(N)}$ whose spectral properties converge to those of T_G . The Kato–Osborn theorem guarantees that for sufficiently large N , the presence of eigenvalue 1 in $T_G^{(N)}$ is equivalent to the existence of a Hamiltonian cycle. This compression is the essential step towards constructing a SAT instance and applying the D-RSP algorithm. The next article (XXXVI.3) establishes a lower bound on the spectral gap of T_G .

References

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